

cuda-quantum docker cpu have different result with gpu

<https://github.com/NVIDIA/cuda-quantum/issues/583>

ROW 12

cuda-quantum docker cpu have different result with gpu #583

New issue



✔ Closed



Guogggg opened on Aug 29, 2023 · edited by Guogggg

Required prerequisites

- ☒ Make sure you've read the [documentation](#). Your issue may be addressed there.
- ☒ Search the [issue tracker](#) to verify that this hasn't already been reported. +1 or comment there if it has.
- ☐ If possible, make a PR with a failing test to give us a starting point to work on!

Describe the bug

```
`import os
import numpy as np
```

```
import cudaq
from cudaq import chemistry

geometry = [('Li', (0, 0, 0)), ('H', (0, 0, 1.5))]
molecule, data = chemistry.create_molecular_hamiltonian(
    geometry, 'sto-3g', 1, 0)
cudaq.set_target('nvidia')
numElectrons = data.n_electrons
numQubits = 2 * data.n_orbitals

kernel, thetas = cudaq.make_kernel([list]
qubits = kernel.qalloc(12)

kernel.x(qubits[0])
kernel.x(qubits[1])
kernel.x(qubits[2])
kernel.x(qubits[3])

cudaq.kernels.uccsd(kernel, qubits, thetas, numElectrons, numQubits)

num_parameters = cudaq.kernels.uccsd_num_parameters(numElectrons, numQubits)
print("num_parameters: ", num_parameters)
optimizer = cudaq.optimizers.SPSA()
energy, params = cudaq.vqe(kernel,
molecule,
optimizer,
parameter_count=num_parameters)
print(energy, params)
```

```
When I comment out cuda.set_device('nvidia'), the result is: -4.703793
```

```
= -4.758690  
= -7.863358  
= -4.931899  
= -4.906214  
= -7.863358  
  
-7.863357621533178 [-1.17716665170367e-12, -1.17716665170367e-12, 1.17716665170367e-12, 1.17716665170367e-12,  
-3.5484976526133087e-12, -1.17716665170367e-12, 3.5484976526133087e-12, 3.5484976526133087e-12,  
3.5484976526133087e-12, -3.5484976526133087e-12, -3.5484976526133087e-12, -1.17716665170367e-12,  
-1.17716665170367e-12, 1.17716665170367e-12, 1.17716665170367e-12, 1.17716665170367e-12, 3.5484976526133087e-12,  
1.17716665170367e-12, -1.17716665170367e-12, 1.17716665170367e-12, -1.17716665170367e-12, 3.5484976526133087e-12,  
3.5484976526133087e-12, -3.5484976526133087e-12, -1.17716665170367e-12, -1.17716665170367e-12,  
3.5484976526133087e-12, 1.17716665170367e-12, 1.17716665170367e-12, -3.5484976526133087e-12,  
3.5484976526133087e-12, -1.17716665170367e-12, -3.5484976526133087e-12, 1.17716665170367e-12,  
-3.5484976526133087e-12, 1.17716665170367e-12, 3.5484976526133087e-12, 1.17716665170367e-12,  
-1.17716665170367e-12, -3.5484976526133087e-12, 3.5484976526133087e-12, -3.5484976526133087e-12,  
-3.5484976526133087e-12, -1.17716665170367e-12, 3.5484976526133087e-12, 1.17716665170367e-12,  
3.5484976526133087e-12, -1.17716665170367e-12, -3.5484976526133087e-12, -1.17716665170367e-12,  
3.5484976526133087e-12, 1.17716665170367e-12, -1.17716665170367e-12, 1.17716665170367e-12,
```

Assignees

1tnguyen

 anthony-santana

bmhowe23

Labels

bug ecosystem

Type

No type

Projects

No projects

Milestone

📌 release 0.5.0

Closed on Jan 22, 2024. 100% complete

Relationships

None yet

Development

No branches or pull requests

Notifications

Customize

 Subscribe

Participants

 +1

Zotero

amccaskey assigned [anthony-santana](#), [itnguyen](#) and [bmhowe23](#) on Aug 29, 2023



bmhowe23 on Aug 29, 2023

Collaborator ...

What GPU are you using? I tried to reproduce the problem on both an A100 and A6000 and I was unable to reproduce the different behavior when toggling whether or not the nvidia target was being used.



Guogggg on Aug 30, 2023

Author ...

您使用什么 GPU? 我尝试在 A100 和 A6000 上重现该问题, 但在切换是否使用 nvidia 目标时无法重现不同的行为。

I am using Tesla T4 and A100 GPUs with CUDA version 12.2 and nvcc version 12.2. When I use the CPU to calculate the energy of LiH, the computed result is very close to the accurate value. However, when I use the GPU, the computed result differs significantly.



Guogggg on Aug 30, 2023

Author ...

您使用什么 GPU? 我尝试在 A100 和 A6000 上恢复该问题, 但在切换是否使用 nvidia 目标时无法恢复不同的行为。

this is demo_code

```
import os
import numpy as np

import cudaq
from cudaq import chemistry
import time
geometry = [('Li', (0., 0., 0.)), ('H', (0., 0., 1.5))]
molecule, data = chemistry.create_molecular_hamiltonian(
    geometry, 'sto-3g', 1, 0)
#cudaq.set_target('nvidia')
# Get the number of fermions and orbitals / qubits
numElectrons = data.n_electrons
numQubits = 2 * data.n_orbitals

# create the ansatz
kernel, thetas = cudaq.make_kernel(list)
#cudaq.set_target('nvidia')
qubits = kernel.qalloc(numQubits)
# hartree fock
# for i in range(numQubits):
#     kernel.x(qubits[i])
kernel.x(qubits[0])
kernel.x(qubits[1])
kernel.x(qubits[2])
kernel.x(qubits[3])

cudaq.kernels.uccsd(kernel, qubits, thetas, numElectrons, numQubits, (variable) numQubits: Any

num_parameters = cudaq.kernels.uccsd_num_parameters(numElectrons, numQubits)
print("num_parameters: ", num_parameters)
# Run VQE
optimizer = cudaq.optimizers.LBFGS()
st = time.time()
energy, params = cudaq.vqe(kernel,
                           molecule,
                           optimizer,
                           parameter_count=num_parameters)
print(energy, params)
en = time.time()

print("time used: ", en-st)
#assert np.isclose(-1.137, energy, rtol=1e-3)
```

and this is cpu result

```
root@7166a5201e9f:/home/cudaq/examples/python# python test_lih.py
/usr/local/lib/python3.10/dist-packages/pyscf/dft/libxc.py:772: UserWarning: Since PySCF-2.3, B3LYP (and B3P86) are changed to the VWN-RPA variant, the same to the B3LYP functional in Gaussian and ORCA (issue 1480). To restore the VWN5 definition, you can put the setting "B3LYP_WITH_VWN5 = True" in pyscf_conf.py
  warnings.warn('Since PySCF-2.3, B3LYP (and B3P86) are changed to the VWN-RPA variant, '
num_parameters: 92
chi = -11.703793
```


Hi @Guogggg,

I cannot reproduce the bug on my system with A100 GPU.

My steps:

- Docker:

```
docker run --gpus all -it --rm nvcr.io/nvidia/cuda-quantum:0.4.0
```

- Run the above example (`nvidia` target):

```
python3 -m pip install openfermionpyscf
python3 test.py
```

- Result:

```
cudaq@9ff453424dea0:~$ python3 test.py
/home/cudaq/.local/lib/python3.10/site-packages/pyscf/dft/libxc.py:772: UserWarning: Since PySCF-2.3, B3LYP (and B3P86) are changed to the VWN-RPA variant, 'B3LYP' will be replaced by 'VWN-RPA' in the future. Please update your input files accordingly.
  warnings.warn('Since PySCF-2.3, B3LYP (and B3P86) are changed to the VWN-RPA variant, '
num_parameters: 92
<H> = -4.703681
<H> = -4.758564
<H> = -7.862600
<H> = -4.931717
<H> = -4.906044
<H> = -7.862600
-7.862599938916624 [-1.1771209396372758e-12, -1.1771209396372758e-12, 1.1771209396372758e-12, 1.17712093963727
```



你好@Guogggg,

我无法在使用 A100 GPU 的系统上重现该错误。

我的步骤：

- 码头工人:

```
docker run --gpus all -it --rm nvcr.io/nvidia/cuda-quantum:0.4.0
```

- 运行上面的例子（`nvidia` 目标）：

```
python3 -m pip install openfermionpyscf
python3 test.py
```

- 结果:

```
cudaq@9f453424dea0:~$ python3 test.py
/home/cudaq/.local/lib/python3.10/site-packages/pyscf/dft/libxc.py:772: UserWarning: Since PySCF-2.5, B3LYP
warnings.warn('Since PySCF-2.3, B3LYP (and B3P86) are changed to the VWN-RPA variant, ')
num_parameters: 92
<H> = -4.703681
<H> = -4.758564
<H> = -7.862600
<H> = -4.931717
<H> = -4.906044
<H> = -7.862600
-7.862599938916624 [-1.1771209396372758e-12, -1.1771209396372758e-12, 1.1771209396372758e-12, 1.177120939
```



```
docker run --gpus all -it --rm nvcr.io/nvidia/cuda-quantum:0.4.0
```

- 运行上面的例子 (`nvidia` 目标) :

```
python3 -m pip install openfermionpyscf
python3 test.py
```

- 结果:

```
cudaq@9f453424dea0:~$ python3 test.py
/home/cudaq/.local/lib/python3.10/site-packages/pyscf/dft/libxc.py:772: UserWarning: Since PySCF-2.3,
warnings.warn('Since PySCF-2.3, B3LYP (and B3P86) are changed to the VWN-RPA variant, '
num_parameters: 92
<H> = -4.703681
<H> = -4.758564
<H> = -7.862600
<H> = -4.931717
<H> = -4.906044
<H> = -7.862600
-7.862599938916624 [-1.1771209396372758e-12, -1.1771209396372758e-12, 1.1771209396372758e-12, 1.1771209396372758e-12]
```

What version of cuda and nvcc are you using?

Need to download custatevec using pip?



bmhowe23 on Aug 30, 2023

Collaborator ...

Like [@1tnguyen](#), I didn't need to install custatevec in order to run this test. Also, nvcc is not installed in the docker container image because it is not needed unless you are modifying cuda-quantum source code, but `nvidia-smi` (which is installed in the container and uses the underlying driver on the host system) reports that it is using CUDA version 12.2 on one of my systems and 12.0 on the other.



Guogggg on Aug 31, 2023

Author ...

Like [@1tnguyen](#), I didn't need to install custatevec in order to run this test. Also, nvcc is not installed in the docker container image because it is not needed unless you are modifying cuda-quantum source code, but `nvidia-smi` (which is installed in the container and uses the underlying driver on the host system) reports that it is using CUDA version 12.2 on one of my systems and 12.0 on the other.

docker:

```
docker run --gpus all -it -u root --rm nvcr.io/nvidia/cuda-quantum:0.4.0
```

```
cd /home/cudaq/examples/python
```

For any example, as long as I add `cudaq.set_target("nvidia")`, the calculation results are incorrect.



[bettinaheim](#) mentioned this on Sep 8, 2023

[Gradients example doesn't make progress on the GPU runners #622](#)



[1tnguyen](#) mentioned this on Sep 27, 2023

[Support Volta+ arch in custatevec backend #696](#)



schweitzpgi on Oct 19, 2023

Collaborator ...

Claims to be fixed.

Microsoft

installed in the container and uses the underlying driver on the host system) reports that it is using CUDA version 12.2 on one of my systems and 12.0 on the other.

docker:

```
docker run --gpus all -it -u root --rm nvcr.io/nvidia/cuda-quantum:0.4.0
```

```
cd /home/cudaq/examples/python
```

For any example, as long as I add `cudaq.set_target("nvidia")`, the calculation results are incorrect.



bettinaheim mentioned this on Sep 8, 2023

✓ [Gradients example doesn't make progress on the GPU runners #622](#)

1tnguyen mentioned this on Sep 27, 2023

🔗 [Support Volta+ arch in custatevec backend #696](#)



schweitzpgi on Oct 19, 2023

Collaborator ...

Claims to be fixed.



schweitzpgi added **verify and close** on Oct 19, 2023



1tnguyen on Oct 20, 2023

Collaborator ...

@Guogggg We were able to reproduce this issue with V100 GPUs and fixed it in [#696](#).

I'm closing this issue now but please feel free to file another if it doesn't work for you. Thanks for reporting the issue.



🔒 **1tnguyen** closed this as **completed** on Oct 20, 2023

bettinaheim added **bug fix** **bug** **ecosystem** and removed **verify and close** **bug fix** on Oct 26, 2023

bettinaheim added this to the [release 0.5.0](#) milestone on Oct 26, 2023

sacpis mentioned this on Jul 25, 2024

✓ [IQM Tests are failing in main #2010](#)